DIAGRAM 1 — S1223 SIDE PROFILE: ALL SPARS, D-BOX, RIB TAIL, GURNEY FLAP, TE ROD

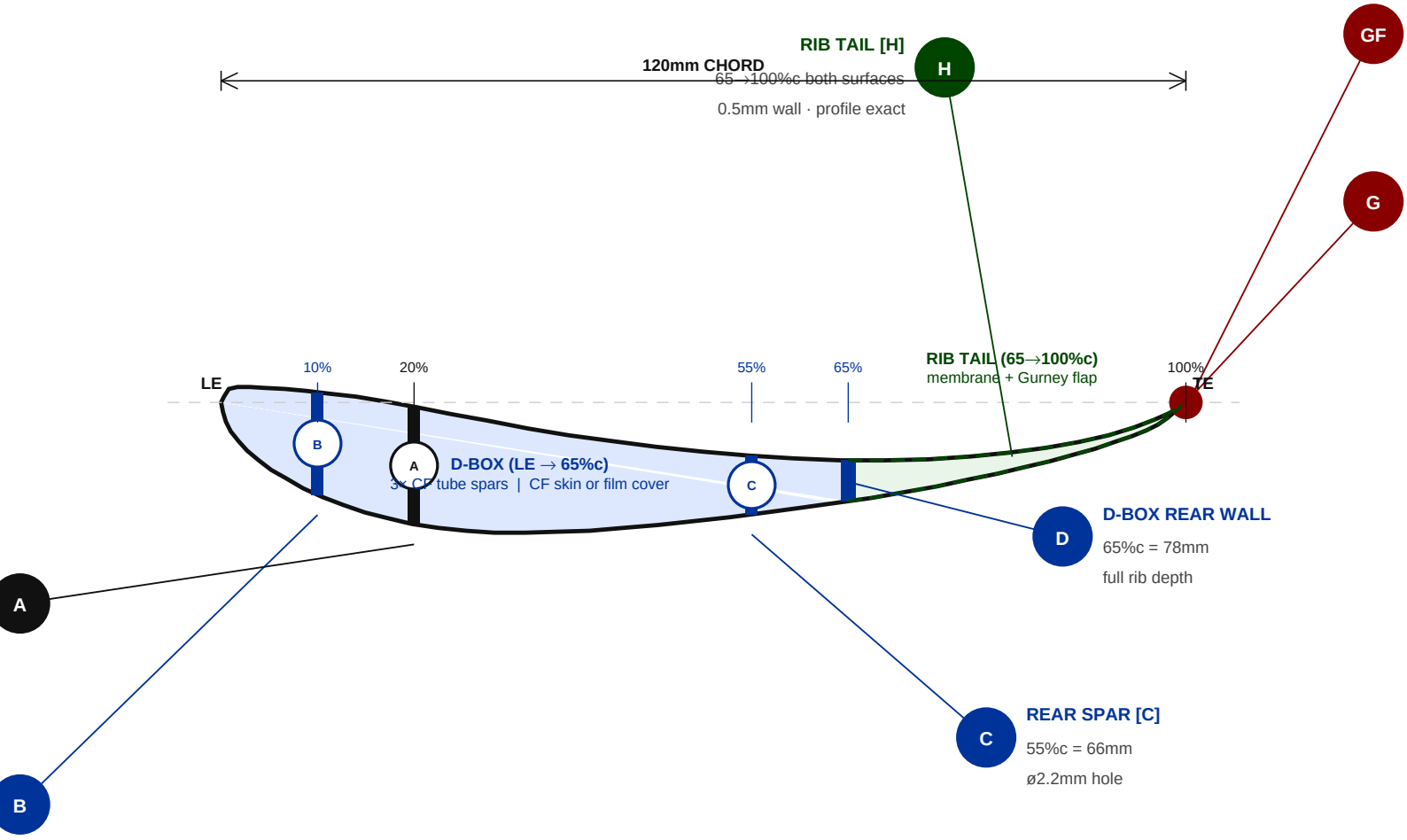
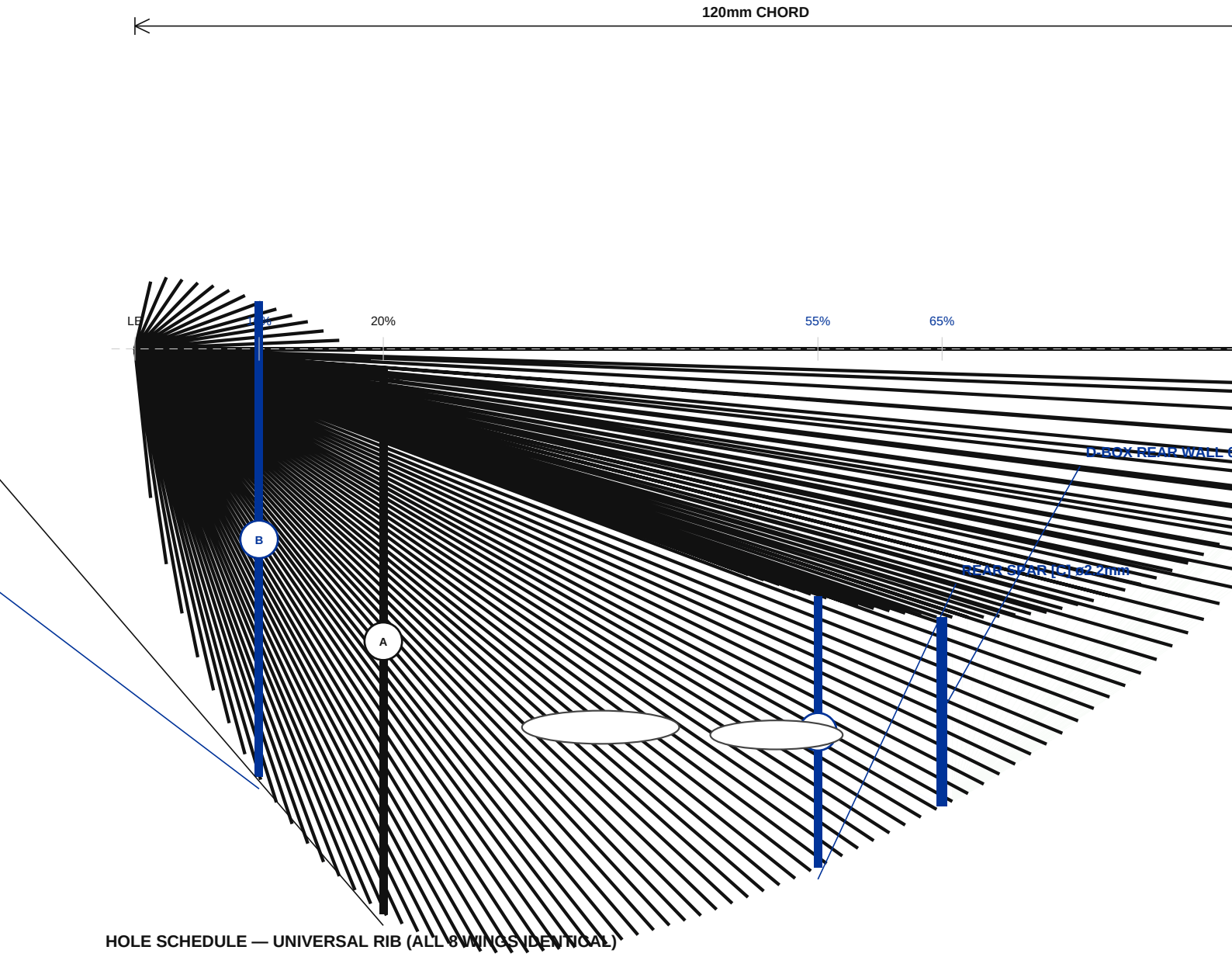
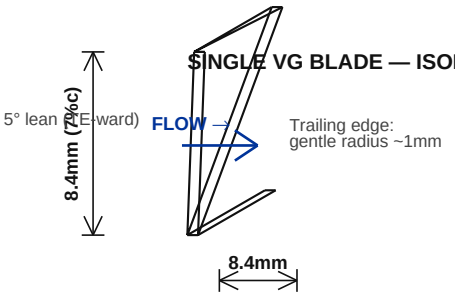


DIAGRAM 2 — RIB FACE (SPAN-DIRECTION VIEW)



HOLE SCHEDULE — UNIVERSAL RIB (ALL 8 WINGS IDENTICAL)

DIAGRAM 3 — VG BLADE GEOMETRY AND MOUNT



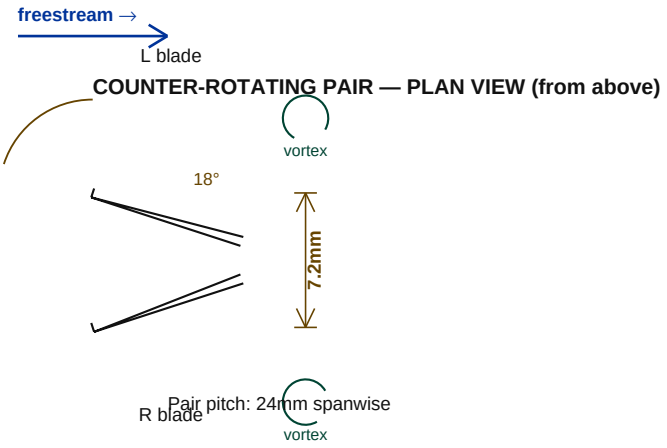
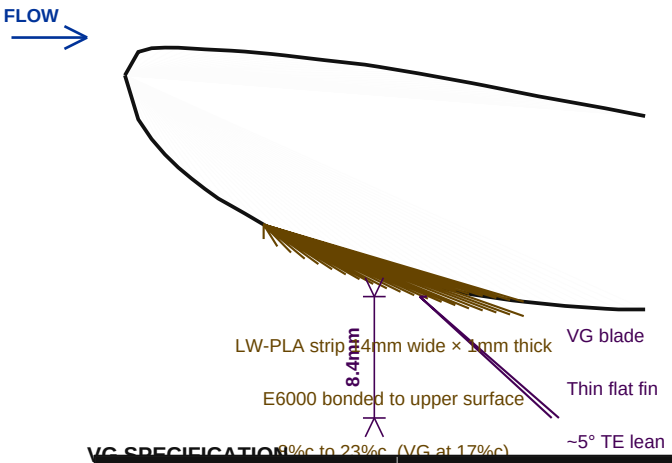


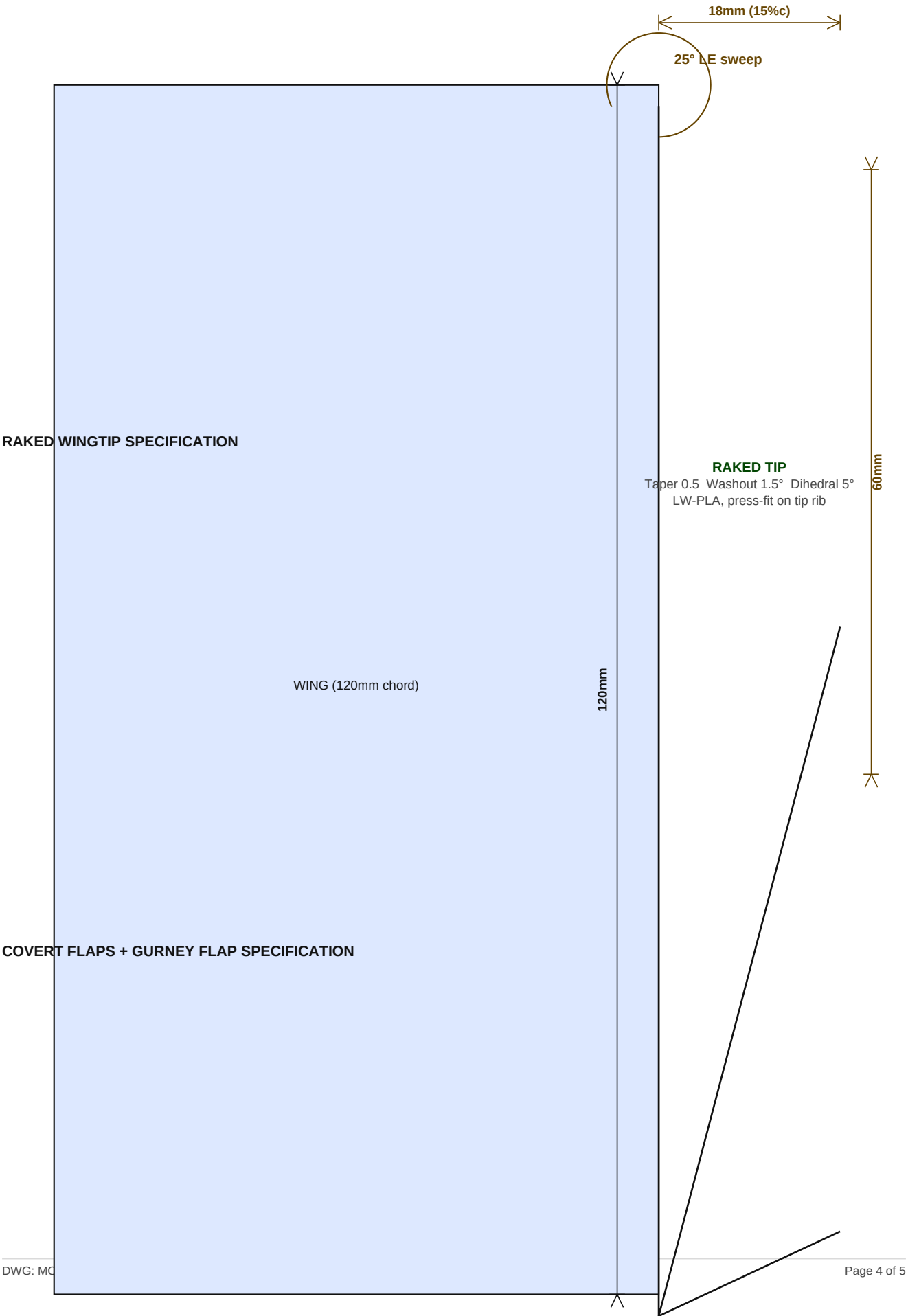
DIAGRAM 3c — MOUNTING STRIP ON UPPER SURFACE (SIDE VIEW)



VG SPECIFICATION

VG type	Thin flat rectangular blade (Micro AeroDynamics style)	Same profile as aviation VGs. Vertical flat fin, 5° TE lean.
Position	17%c = 20.4mm from LE, upper surface	Selig 1997 S1223 wind tunnel (CL,max 2.20→2.30)
Height h	8.4mm (7%c)	5-8%c range; h ≈ 3.5× BL thickness at Re 4,000-6,000
Chord-length	8.4mm (7%c)	Matched to height — standard VG proportioning
Thickness	0.5mm (LW-PLA, 2 perimeters)	Minimum for structural integrity at this height
Toe angle	18° to freestream	Optimal 15-20° range; good vortex/drag balance
Lean	~5° toward trailing edge	Mirrors aviation VG geometry; aids vortex attachment
TE profile	Gentle radius ~1mm, not sharp	Reduces drag vs sharp TE; matches full-scale VG shape
Pair gap	7.2mm (6%c) fin-to-fin	Counter-rotating pair; common-flow-down configuration
Pair pitch	24mm spanwise (20%c)	~3× height; one pair per rib bay
Mounting strip	LW-PLA, 14mm chord × full span × 1mm base	Single print per wing; all 8 identical

DIAGRAM 4 — RAKED WINGTIP PLANFORM AND CROSS-SECTION



MCO-1D WING — CONSOLIDATED SPECIFICATION (ALL CONFIRMED DECISIONS)